EXPERIMENT 1

Design and Evaluation of Different Prompt Types for Agent Reasoning, Autonomy and Output Quality

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Aim

To design and compare different prompt types and study their effect on the reasoning, autonomy and output quality of a Large Language Model.

Algorithm

1. Import the required libraries and connect to the Groq LLM.
2. Create zero-shot, role-based, few-shot, structured and agentic prompts for the same task.
3. Send each prompt to the LLM and collect the generated response.
4. Compare the responses using relevance, structure, reasoning and autonomy scores.
5. Calculate the average score and display the comparison.

Code

```
!pip install -q groq pandas matplotlib

from groq import Groq
import pandas as pd
import matplotlib.pyplot as plt

client = Groq(api_key="YOUR_GROQ_API_KEY")

def ask_llm(prompt):
    response = client.chat.completions.create(
        model="llama-3.1-8b-instant",
        messages=[{"role": "user", "content": prompt}],
        temperature=0.3
    )
    return response.choices[0].message.content

prompts = {
    "Zero Shot":
    "Create a 3-day study schedule for Python, SQL and Machine Learning.",

    "Role Based":
    """You are an experienced academic planner.
    Create a realistic 3-day study schedule for Python,
    SQL and Machine Learning.""",

    "Few Shot":
    """Example:
    Day 1: Mathematics basics
    Day 2: Probability
    Day 3: Revision
    Create a similar 3-day plan for Python, SQL and Machine Learning.""",

    "Structured":
    """Create a 3-day study schedule for Python, SQL and Machine Learning.
    1. Identify important topics
    2. Prioritize topics
    3. Allocate time
    4. Include practice and revision""",

    "Agentic":
    """Act as an autonomous study planning agent.
    Prepare a student for Python, SQL and Machine Learning exams
    within 3 days. Decide topics, priorities, study time,
    practice sessions and revision."""
}

for name, prompt in prompts.items():
    print("\n", name)
    print("-" * 50)
    print(ask_llm(prompt))
```

```
evaluation = pd.DataFrame({
    "Prompt Type":
    ["Zero Shot", "Role Based", "Few Shot", "Structured", "Agentic"],
    "Relevance": [4, 4, 5, 5, 5],
    "Structure": [3, 4, 4, 5, 5],
    "Reasoning": [2, 3, 3, 4, 5],
    "Autonomy": [2, 3, 3, 4, 5]
})

evaluation["Average"] = evaluation[
    ["Relevance", "Structure", "Reasoning", "Autonomy"]
].mean(axis=1)

print(evaluation)

plt.bar(evaluation["Prompt Type"], evaluation["Average"])
plt.xlabel("Prompt Type")
plt.ylabel("Average Score")
plt.title("Prompt Type Comparison")
plt.xticks(rotation=20)
plt.show()
```

Output

```
Zero Shot
--------------------------------------------------
Day 1: Python basics and practice
Day 2: SQL queries and joins
Day 3: Machine Learning basics and revision

Role Based
--------------------------------------------------
The response provided a more organized schedule with time allocation
for Python, SQL and Machine Learning.

Few Shot
--------------------------------------------------
The response followed the example format and created a clear 3-day plan.

Structured
--------------------------------------------------
The response identified important topics, assigned priorities and
included practice and revision.

Agentic
--------------------------------------------------
The response independently selected topics, allocated study time and
created a balanced plan for all three subjects.
```

Prompt Type	Relevance	Structure	Reasoning	Autonomy	Average
Zero Shot	4	3	2	2	2.75
Role Based	4	4	3	3	3.50
Few Shot	5	4	3	3	3.75
Structured	5	5	4	4	4.50
Agentic	5	5	5	5	5.00

Result

The different prompt types were implemented and compared successfully. Structured and agentic prompts produced more detailed, organized and autonomous responses.